Quantium_Task2_R_Code_Analysis

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————————— 1. LOAD PACKAGES —————————

```
library(data.table)
```

```
## Warning: package 'data.table' was built under R version 4.5.3
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.5.3

##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:data.table':
##
##     hour, isoweek, isoyear, mday, minute, month, quarter, second, wday,
##     week, yday, year

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

Global theme

```
theme_set(theme_bw())
theme_update(plot.title = element_text(hjust = 0.5))
```

———————2. LOAD & PREPARE DATA ———————

```r
data <- fread("QVI_merged_data.csv")
data[, YEARMONTH := year(DATE) * 100 + month(DATE)]

measureOverTime <- data[, .(
  totSales = sum(TOT_SALES),
  nCustomers = uniqueN(LYLTY_CARD_NBR),
  nTxnPerCust = uniqueN(TXN_ID) / uniqueN(LYLTY_CARD_NBR),
  nChipsPerTxn = sum(PROD_QTY) / uniqueN(TXN_ID),
  avgPricePerUnit = sum(TOT_SALES) / sum(PROD_QTY)),
  by = .(STORE_NBR, YEARMONTH)][order(STORE_NBR, YEARMONTH)]

storesWithFullObs <- measureOverTime[, .N, by = STORE_NBR][N == 12, STORE_NBR]
preTrialMeasures <- measureOverTime[YEARMONTH < 201902
                                    & STORE_NBR %in% storesWithFullObs,]
```

3. SIMILARITY FUNCTIONS

```r
calculateCorrelation <- function(inputTable, metricCol, storeComparison) {
  calcCorrTable <- data.table(Store1 = numeric(),
                              Store2 = numeric(), corr_measure = numeric())
  storeNumbers <- unique(inputTable[, STORE_NBR])
  for (i in storeNumbers) {
      calculatedMeasure <- data.table(
          Store1 = storeComparison,
          Store2 = i,
          corr_measure = cor(inputTable[STORE_NBR == storeComparison, eval(metricCol)],
                             inputTable[STORE_NBR == i, eval(metricCol)])
      )
      calcCorrTable <- rbind(calcCorrTable, calculatedMeasure)
  }
  return(calcCorrTable)
}

calculateMagnitudeDistance <- function(inputTable, metricCol, storeComparison) {
     calcDistTable <- data.table(Store1 = numeric(), Store2 = numeric(),
                                 YEARMONTH = numeric(), measure = numeric())
     storeNumbers <- unique(inputTable[, STORE_NBR])
     for (i in storeNumbers) {
         calculatedMeasure <- data.table(
             Store1 = storeComparison,
             Store2 = i,
             YEARMONTH = inputTable[STORE_NBR == storeComparison, YEARMONTH],
             measure = abs(inputTable[STORE_NBR == storeComparison, eval(metricCol)] -
                           inputTable[STORE_NBR == i, eval(metricCol)])
        )
        calcDistTable <- rbind(calcDistTable, calculatedMeasure)
    }
    minMaxDist <- calcDistTable[, .(minDist = min(measure),
                                    maxDist = max(measure)), by = .(Store1, YEARMONTH)]
    distTable <- merge(calcDistTable, minMaxDist, by = c("Store1", "YEARMONTH"))
```

```
    distTable[, magnitudeMeasure := 1 - (measure - minDist) / (maxDist - minDist)]
    return(distTable[, .(mag_measure = mean(magnitudeMeasure)), by = .(Store1, Store2)])
}

get_control_store <- function(trial_store_nbr) {
     corr_sales <- calculateCorrelation(preTrialMeasures,
                                        quote(totSales), trial_store_nbr)
     corr_cust <- calculateCorrelation(preTrialMeasures,
                                       quote(nCustomers), trial_store_nbr)
     mag_sales <- calculateMagnitudeDistance(preTrialMeasures,
                                             quote(totSales), trial_store_nbr)
     mag_cust <- calculateMagnitudeDistance(preTrialMeasures,
                                            quote(nCustomers), trial_store_nbr)
     scores <- merge(corr_sales, mag_sales, by = c("Store1", "Store2"))
     scores[, scoreSales := corr_measure * 0.5 + mag_measure * 0.5]
     scores_cust <- merge(corr_cust, mag_cust, by = c("Store1", "Store2"))
     scores_cust[, scoreCust := corr_measure * 0.5 + mag_measure * 0.5]
     final_scores <- merge(scores[, .(Store1, Store2, scoreSales)],
                           scores_cust[, .(Store1, Store2, scoreCust)],
                           by = c("Store1", "Store2"))
     final_scores[, finalScore := scoreSales * 0.5 + scoreCust * 0.5]
     return(final_scores[order(-finalScore)][2, Store2])}
```

4. SELECT CONTROL STORES

```
trial_stores <- c(77, 86, 88)
control_stores <- sapply(trial_stores, get_control_store)
selection_summary <- data.table(Trial = trial_stores, Control = control_stores)

cat("\n===== CONTROL STORE MATCHES =====\n")
```

```
##
## ===== CONTROL STORE MATCHES =====
```

```
# ===== CONTROL STORE MATCHES =====
print(selection_summary)
```

```
##    Trial Control
##    <num>   <num>
## 1:    77     233
## 2:    86     155
## 3:    88     237
```

```
##   Trial Control
  ##  <num>   <num>
## 1:    77     233
## 2:    86     155
## 3:    88     237
```

5. ENHANCED ANALYSIS FUNCTION

```r
analyze_trial_store <- function(trial, control) {
     cat(paste0("\n", paste(rep("=", 60), collapse = ""), "\n"))
     cat(paste0("ANALYSIS FOR TRIAL STORE ", trial, " vs CONTROL STORE ", control, "\n"))
     cat(paste(rep("=", 60), collapse = ""), "\n")

     # ---- 5.1 Pre-trial trend plots (sales & customers) ----

     # Sales pre-trend
     preSales <- measureOverTime[YEARMONTH < 201902]
     preSales[, Store_type := ifelse(STORE_NBR == trial, "Trial",
                                     ifelse(STORE_NBR == control,
                                            "Control", "Other stores"))]
     preSalesSum <- preSales[, .(totSales = mean(totSales)),
                              by = .(YEARMONTH, Store_type)]
     preSalesSum[, Date := as.Date(paste(YEARMONTH %/% 100,
                                         YEARMONTH %% 100, 1, sep = "-"))]

     p_pre_sales <- ggplot(preSalesSum, aes(Date,
                                            totSales, color = Store_type)) +
         geom_line(size = 1.2) +
         scale_color_manual(values = c("Trial" = "blue",
                                       "Control" = "darkgreen",
                                       "Other stores" = "grey70")) +
         labs(title = paste0("Pre-Trial Total Sales: Store ",
                              trial, " vs Control ", control),
              y = "Total Sales ($)", x = "Month") +
         theme(legend.position = "bottom")
     ggsave(paste0("pre_trial_sales_store_", trial, ".png"),
            p_pre_sales, width = 10, height = 6, dpi = 300)
     print(p_pre_sales)

     # Customers pre-trend
     preCust <- measureOverTime[YEARMONTH < 201902]
     preCust[, Store_type := ifelse(STORE_NBR == trial, "Trial",
                                    ifelse(STORE_NBR == control,
                                           "Control", "Other stores"))]
     preCustSum <- preCust[, .(nCustomers = mean(nCustomers)),
                            by = .(YEARMONTH, Store_type)]
     preCustSum[, Date := as.Date(paste(YEARMONTH %/% 100,
                                        YEARMONTH %% 100, 1, sep = "-"))]

     p_pre_cust <- ggplot(preCustSum, aes(Date,
                                          nCustomers, color = Store_type)) +
         geom_line(size = 1.2) +
         scale_color_manual(values = c("Trial" = "blue",
                                       "Control" = "darkgreen",
                                       "Other stores" = "grey70")) +
         labs(title = paste0("Pre-Trial Number of Customers: Store ",
                              trial, " vs Control ", control),
              y = "Number of Customers", x = "Month") +
         theme(legend.position = "bottom")
```

```
ggsave(paste0("pre_trial_customers_store_", trial, ".png"),
       p_pre_cust, width = 10, height = 6, dpi = 300)
print(p_pre_cust)

# ---- 5.2 Scale control store ----
scale_sales <- preTrialMeasures[STORE_NBR == trial, sum(totSales)] /
    preTrialMeasures[STORE_NBR == control, sum(totSales)]
scale_cust <- preTrialMeasures[STORE_NBR == trial, sum(nCustomers)] /
    preTrialMeasures[STORE_NBR == control, sum(nCustomers)]

measureSales <- copy(measureOverTime)
control_sales <- measureSales[STORE_NBR == control][, scaledSales
                                                    := totSales * scale_sales]
control_cust <- measureSales[STORE_NBR == control][, scaledCust
                                                   := nCustomers * scale_cust]

# ---- 5.3 Sales uplift assessment ----
comp_sales <- merge(control_sales[, .(YEARMONTH, scaledSales)],
                    measureSales[STORE_NBR == trial,
                                 .(YEARMONTH, totSales)],
                    by = "YEARMONTH")
comp_sales[, pctDiff := (totSales - scaledSales) / scaledSales]
stdDev_sales <- sd(comp_sales[YEARMONTH < 201902, pctDiff])
comp_sales[, tValue := pctDiff / stdDev_sales]

cat("\n----- TOTAL SALES UPLIFT -----\n")
trial_results_sales <- comp_sales[YEARMONTH %in% 201902:201904,
                                  .(YEARMONTH, tValue,
                                    Significant = tValue > qt(0.95, 7))]
print(trial_results_sales)

# ---- 5.4 Customer uplift assessment ----
comp_cust <- merge(control_cust[, .(YEARMONTH, scaledCust)],
                   measureSales[STORE_NBR == trial, .(YEARMONTH, nCustomers)],
                   by = "YEARMONTH")
comp_cust[, pctDiff := (nCustomers - scaledCust) / scaledCust]
stdDev_cust <- sd(comp_cust[YEARMONTH < 201902, pctDiff])
comp_cust[, tValue := pctDiff / stdDev_cust]

cat("\n----- NUMBER OF CUSTOMERS UPLIFT -----\n")
trial_results_cust <- comp_cust[YEARMONTH %in% 201902:201904,
                                .(YEARMONTH, tValue,
                                  Significant = tValue > qt(0.95, 7))]
print(trial_results_cust)

# ---- 5.5 Sales uplift plot with CI ----
plotData_sales <- measureSales[STORE_NBR %in% c(trial, control)]
plotData_sales[, Store_type := ifelse(STORE_NBR == trial, "Trial", "Control")]
plotData_sales[, Date := as.Date(paste(YEARMONTH %/% 100,
                                       YEARMONTH %% 100, 1, sep = "-"))]
plotData_sales_sub <- plotData_sales[, .(Date, totSales, Store_type)]

ci_upper_sales <- plotData_sales[Store_type == "Control",
```

```r
                                .(Date, totSales = totSales
                                  * scale_sales * (1 + stdDev_sales * 2),
                                  Store_type = "Upper CI")]
ci_lower_sales <- plotData_sales[Store_type == "Control",
                                .(Date, totSales = totSales
                                  * scale_sales * (1 - stdDev_sales * 2),
                                  Store_type = "Lower CI")]

combined_sales <- rbind(plotData_sales_sub, ci_upper_sales, ci_lower_sales)

p_sales_uplift <- ggplot(combined_sales, aes(Date, totSales,
                                      color = Store_type, linetype = Store_type)) +
    geom_rect(aes(xmin = as.Date("2019-02-01"),
                  xmax = as.Date("2019-04-01"), ymin = 0, ymax = Inf),
              fill = "grey85", alpha = 0.3, color = NA, show.legend = FALSE) +
    geom_line(size = 1.2) +
    scale_color_manual(values = c("Trial" = "blue", "Control" = "darkgreen",
                                  "Upper CI" = "grey50", "Lower CI" = "grey50")) +
    scale_linetype_manual(values = c("Trial" = "solid", "Control" = "solid",
                                     "Upper CI" = "dashed", "Lower CI" = "dashed")) +
    labs(title = paste0("Store ", trial, " Total Sales vs Control ", control),
         subtitle = "Shaded area: trial period (Feb-Apr 2019)
         . Dashed lines: ±2 SD confidence interval",
         y = "Total Sales ($)", x = "Month") +
    theme(legend.position = "bottom")
ggsave(paste0("uplift_sales_store_", trial, ".png"),
       p_sales_uplift, width = 10, height = 6, dpi = 300)
print(p_sales_uplift)

# ---- 5.6 Customer uplift plot with CI ----
plotData_cust <- measureSales[STORE_NBR %in% c(trial, control)]
plotData_cust[, Store_type := ifelse(STORE_NBR == trial, "Trial", "Control")]
plotData_cust[, Date := as.Date(paste(
  YEARMONTH %/% 100, YEARMONTH %% 100, 1, sep = "-"))]
plotData_cust_sub <- plotData_cust[, .(Date, nCustomers, Store_type)]

ci_upper_cust <- plotData_cust[Store_type == "Control",
                              .(Date, nCustomers = nCustomers * scale_cust
                                * (1 + stdDev_cust * 2),
                                Store_type = "Upper CI")]
ci_lower_cust <- plotData_cust[Store_type == "Control",
                              .(Date, nCustomers = nCustomers * scale_cust
                                * (1 - stdDev_cust * 2),
                                Store_type = "Lower CI")]

combined_cust <- rbind(plotData_cust_sub, ci_upper_cust, ci_lower_cust)

p_cust_uplift <- ggplot(combined_cust, aes(Date, nCustomers,
                                    color = Store_type, linetype = Store_type)) +
    geom_rect(aes(xmin = as.Date("2019-02-01"),
                  xmax = as.Date("2019-04-01"), ymin = 0, ymax = Inf),
              fill = "grey85", alpha = 0.3, color = NA, show.legend = FALSE) +
    geom_line(size = 1.2) +
```

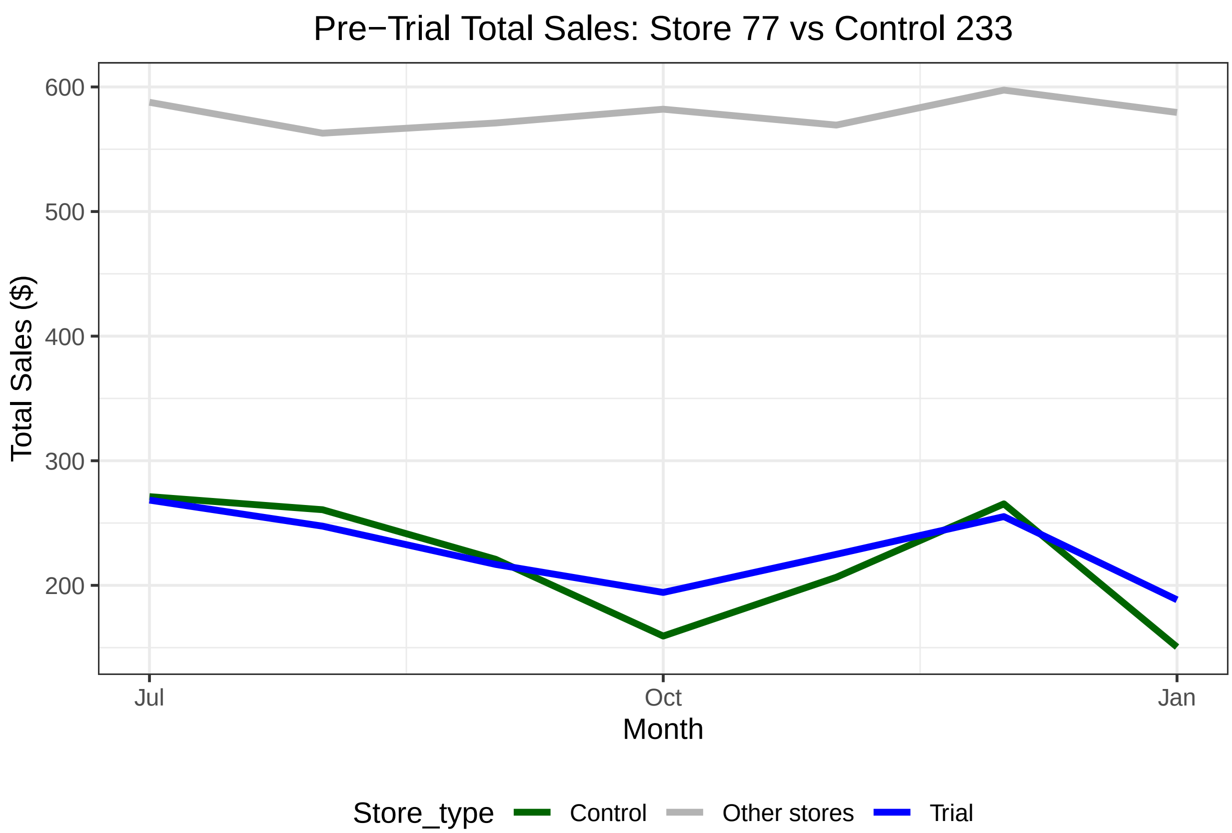
```r
        scale_color_manual(values = c("Trial" = "blue", "Control" = "darkgreen",
                                      "Upper CI" = "grey50", "Lower CI" = "grey50")) +
        scale_linetype_manual(values = c("Trial" = "solid", "Control" = "solid",
                                         "Upper CI" = "dashed", "Lower CI" = "dashed")) +
        labs(title = paste0("Store ", trial, " Number of Customers vs Control ", control),
             subtitle = "Shaded area: trial period (Feb-Apr 2019)
             . Dashed lines: ±2 SD confidence interval",
             y = "Number of Customers", x = "Month") +
        theme(legend.position = "bottom")
    ggsave(paste0("uplift_customers_store_", trial, ".png"),
           p_cust_uplift, width = 10, height = 6, dpi = 300)
    print(p_cust_uplift)

    return(list(sales_sig = sum(trial_results_sales$Significant) >= 2,
                cust_sig = sum(trial_results_cust$Significant) >= 2))}
```

6. RUN ANALYSIS FOR ALL STORES

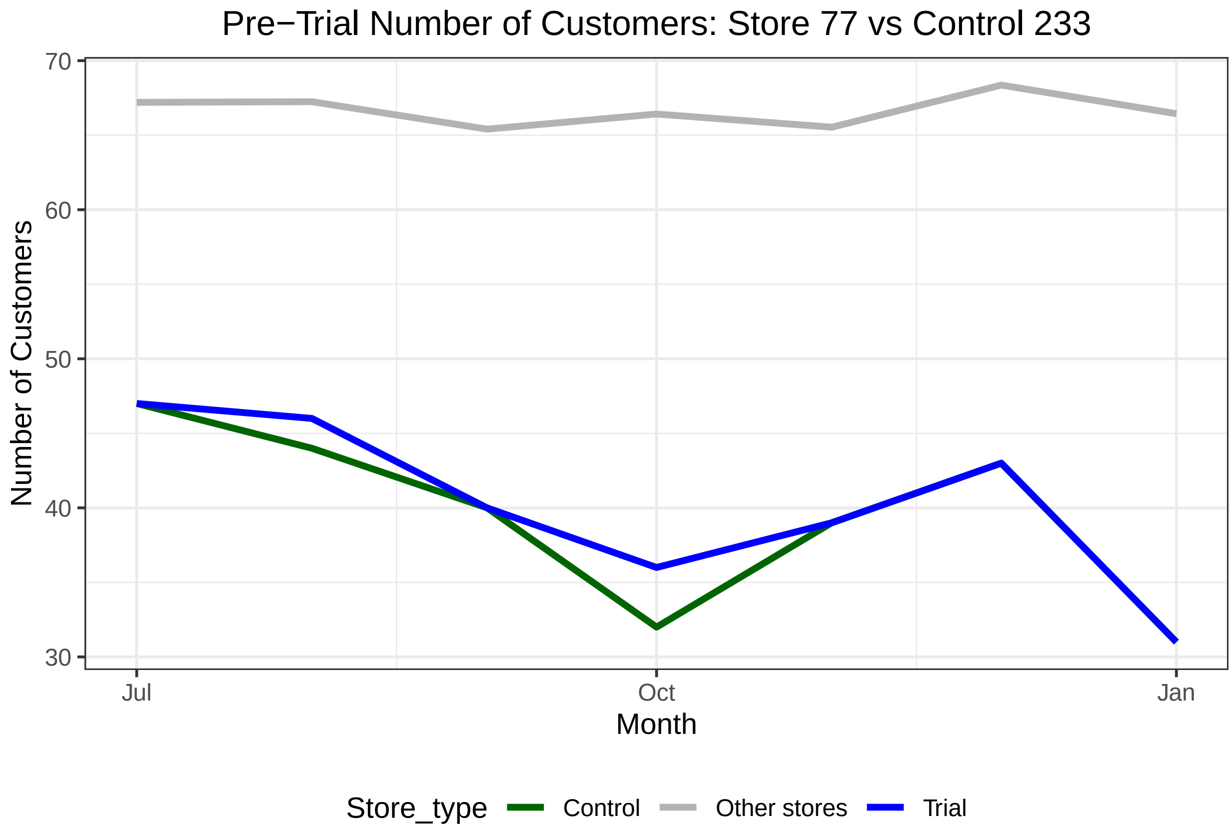
```r
res77 <- analyze_trial_store(77, 233)
```

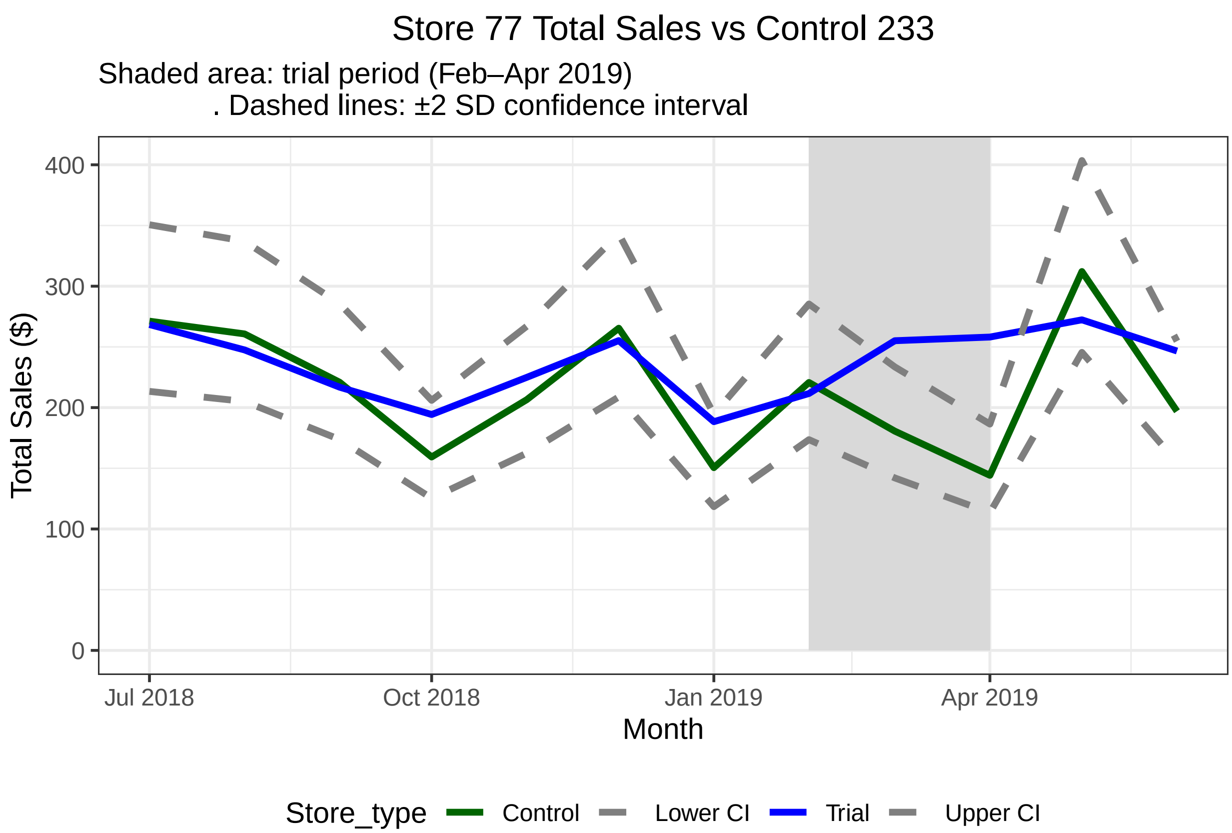
```
##
## ============================================================
## ANALYSIS FOR TRIAL STORE 77 vs CONTROL STORE 233
## ============================================================
```

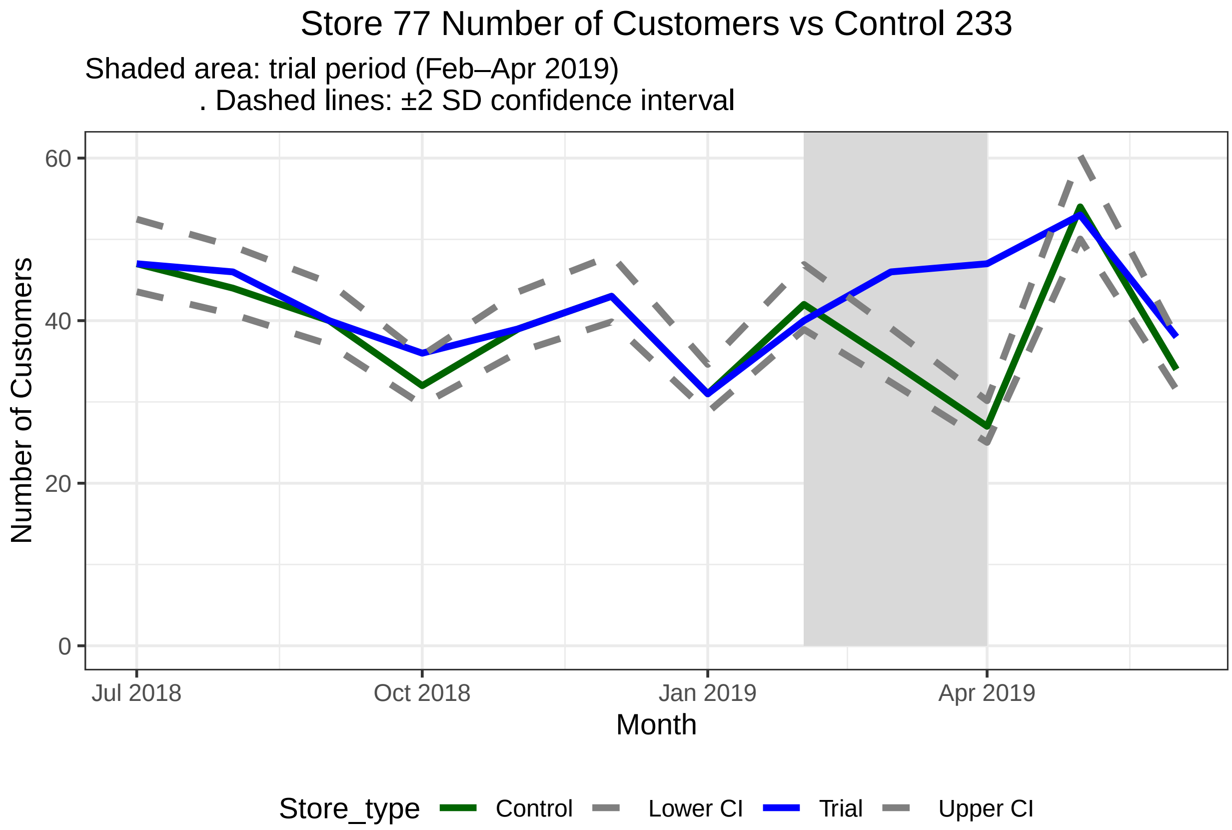
```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
##
## ----- TOTAL SALES UPLIFT -----
## Key: <YEARMONTH>
##    YEARMONTH     tValue Significant
##        <num>      <num>      <lgcl>
## 1:    201902 -0.6398255       FALSE
## 2:    201903  2.9450255        TRUE
## 3:    201904  5.9263924        TRUE
##
## ----- NUMBER OF CUSTOMERS UPLIFT -----
## Key: <YEARMONTH>
##    YEARMONTH    tValue Significant
##        <num>     <num>      <lgcl>
## 1:    201902 -1.460014       FALSE
## 2:    201903  6.158208        TRUE
## 3:    201904 15.135234        TRUE
```

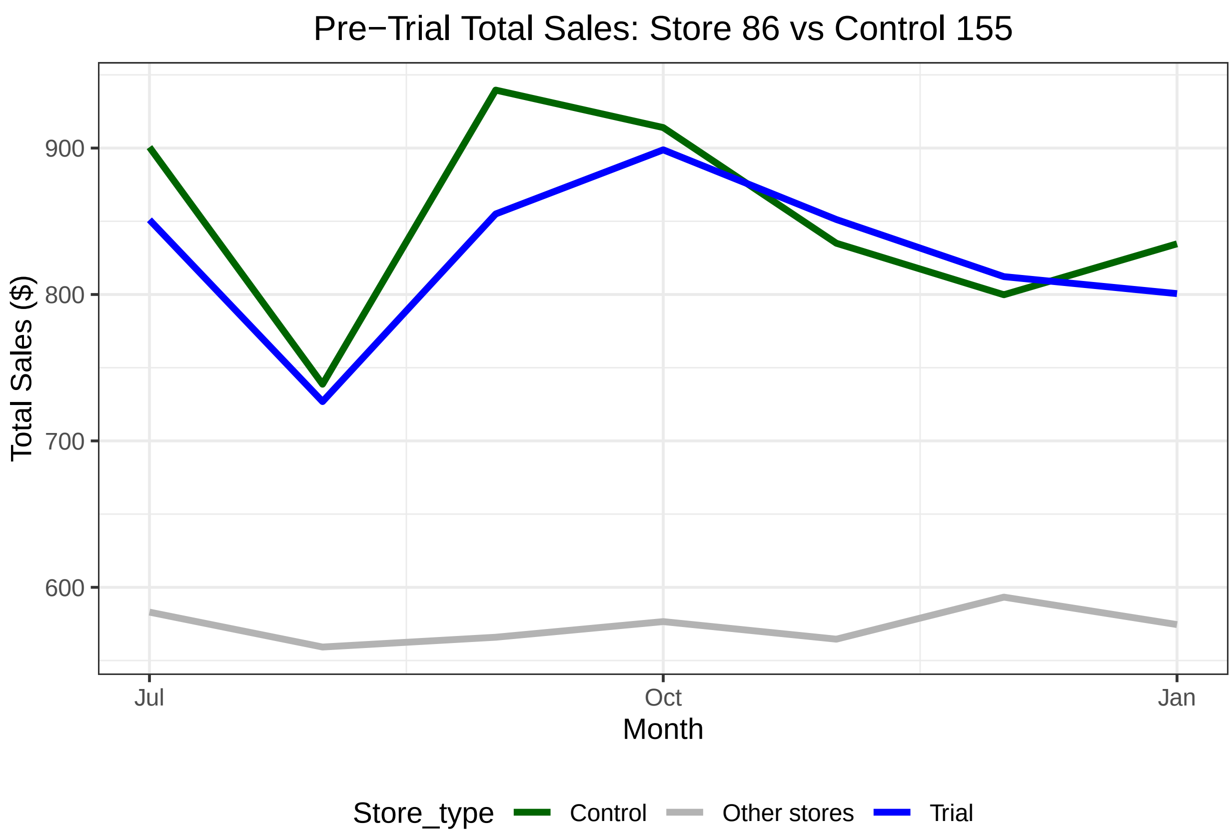

Pre−Trial Number of Customers: Store 77 vs Control 233
Number of Customers
70
60
50
40
30
Jul
Oct
Jan
Month
Store_type
Control
Other stores
Trial


Store 77 Total Sales vs Control 233
Shaded area: trial period (Feb–Apr 2019)
. Dashed lines: ±2 SD confidence interval
Total Sales ($)
400
300
200
100
0
Jul 2018
Oct 2018
Jan 2019
Apr 2019
Month
Store_type
Control
Lower CI
Trial
Upper CI

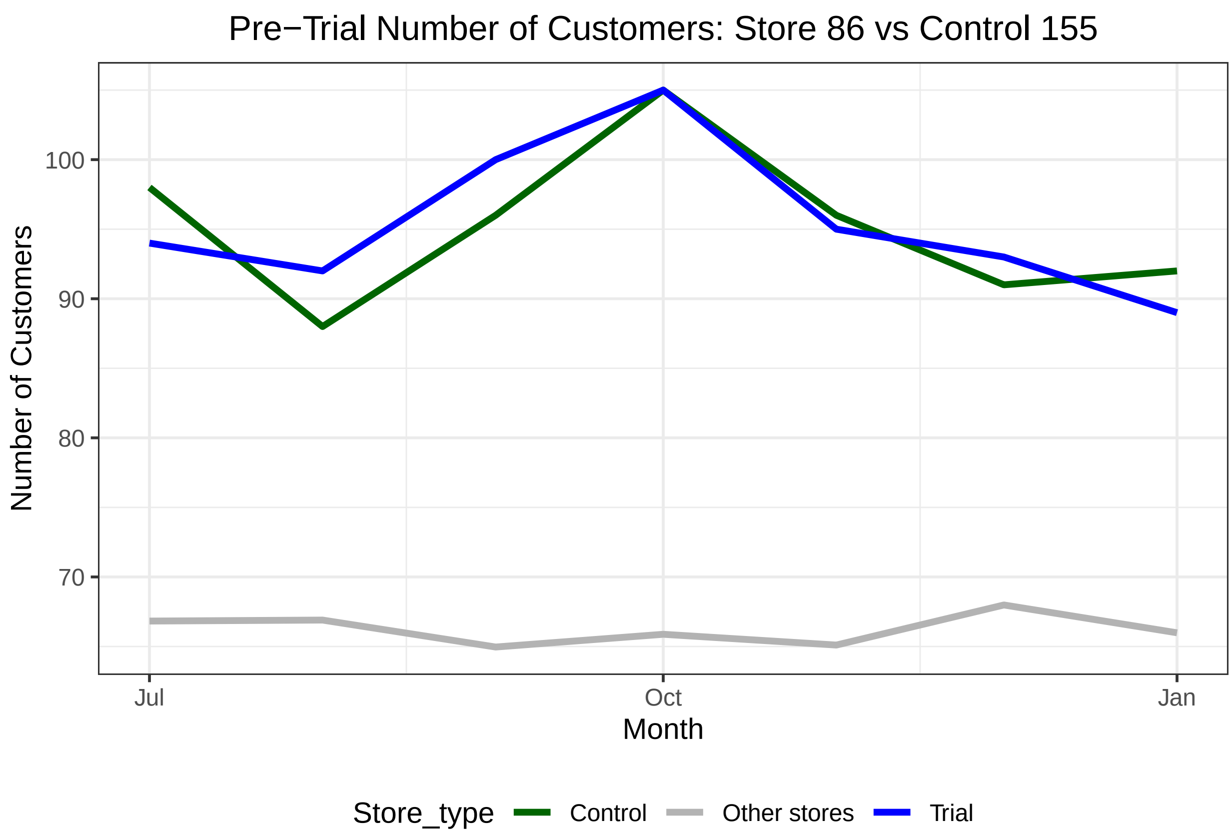


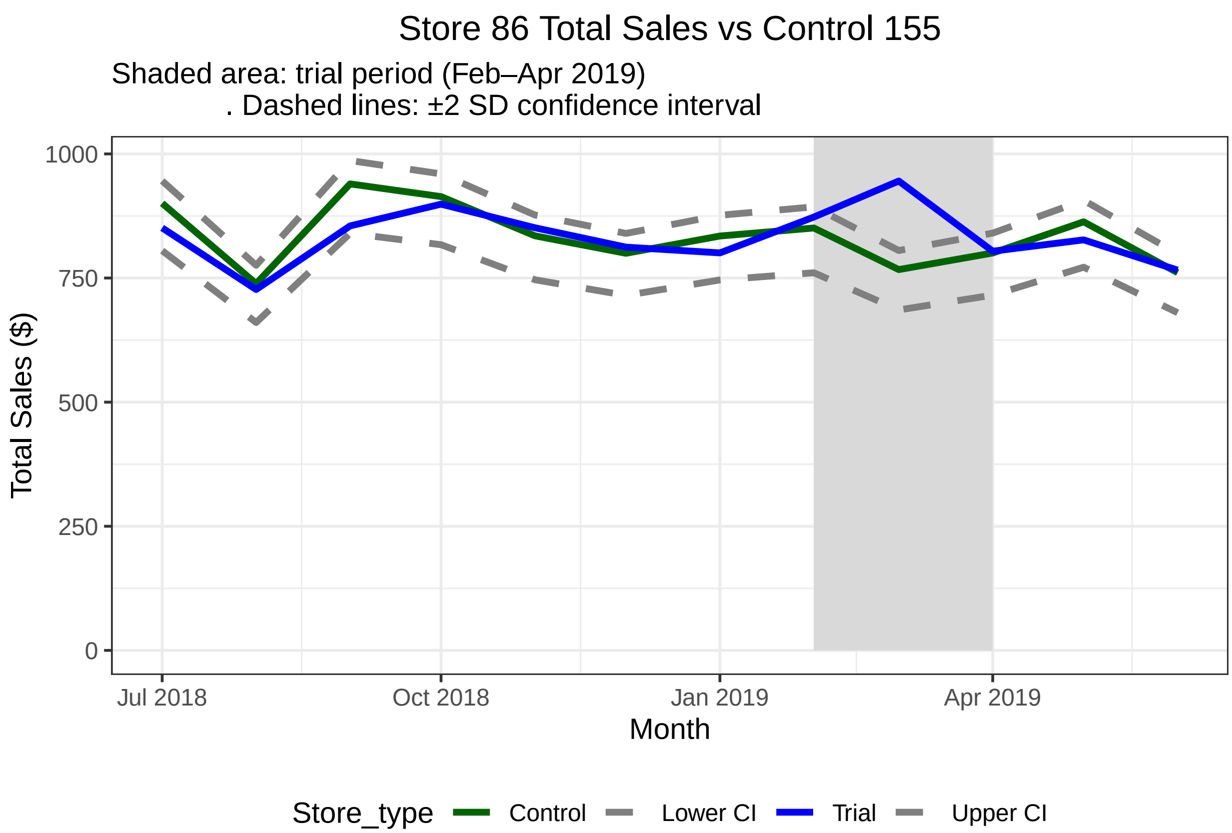

```
# ============================================================
# ANALYSIS FOR TRIAL STORE 77 vs CONTROL STORE 233
# ============================================================

# ----- TOTAL SALES UPLIFT -----
#  Key: <YEARMONTH>
#  YEARMONTH      tValue Significant
#       <num>      <num>      <lgcl>
# 1:     201902 -0.6398255        FALSE
# 2:     201903  2.9450255         TRUE
# 3:     201904  5.9263924         TRUE

# ----- NUMBER OF CUSTOMERS UPLIFT -----
# Key: <YEARMONTH>
 #  YEARMONTH     tValue Significant
 #       <num>     <num>      <lgcl>
# 1:     201902 -1.460014        FALSE
# 2:     201903  6.158208         TRUE
# 3:     201904 15.135234         TRUE
res86 <- analyze_trial_store(86, 155)
```

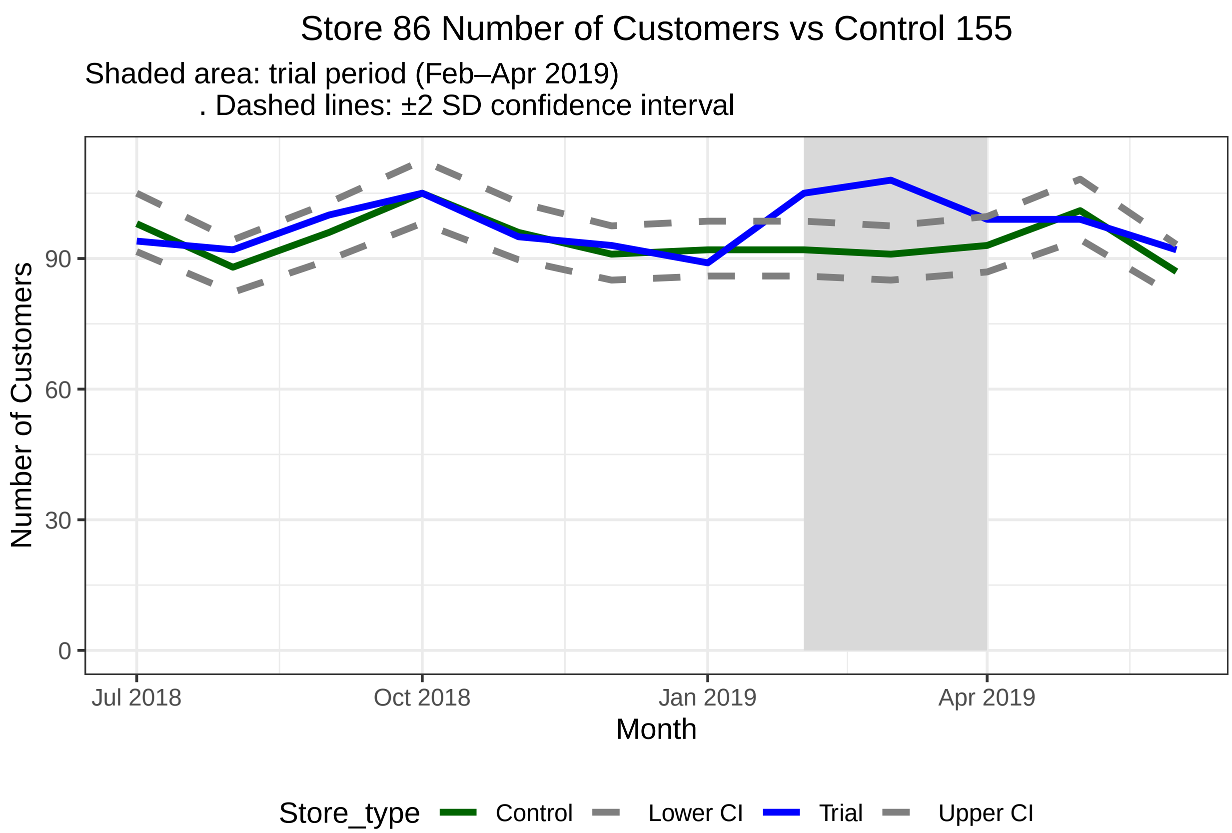
```
##
## ============================================================
## ANALYSIS FOR TRIAL STORE 86 vs CONTROL STORE 155
## ============================================================
```




```
##
## ----- TOTAL SALES UPLIFT -----
## Key: <YEARMONTH>
##    YEARMONTH    tValue Significant
##        <num>     <num>      <lgcl>
## 1:    201902 1.3791146       FALSE
## 2:    201903 6.6777680        TRUE
## 3:    201904 0.8316513       FALSE
##
## ----- NUMBER OF CUSTOMERS UPLIFT -----
## Key: <YEARMONTH>
##    YEARMONTH   tValue Significant
##        <num>    <num>      <lgcl>
## 1:    201902 4.039845        TRUE
## 2:    201903 5.369179        TRUE
## 3:    201904 1.796826       FALSE
```

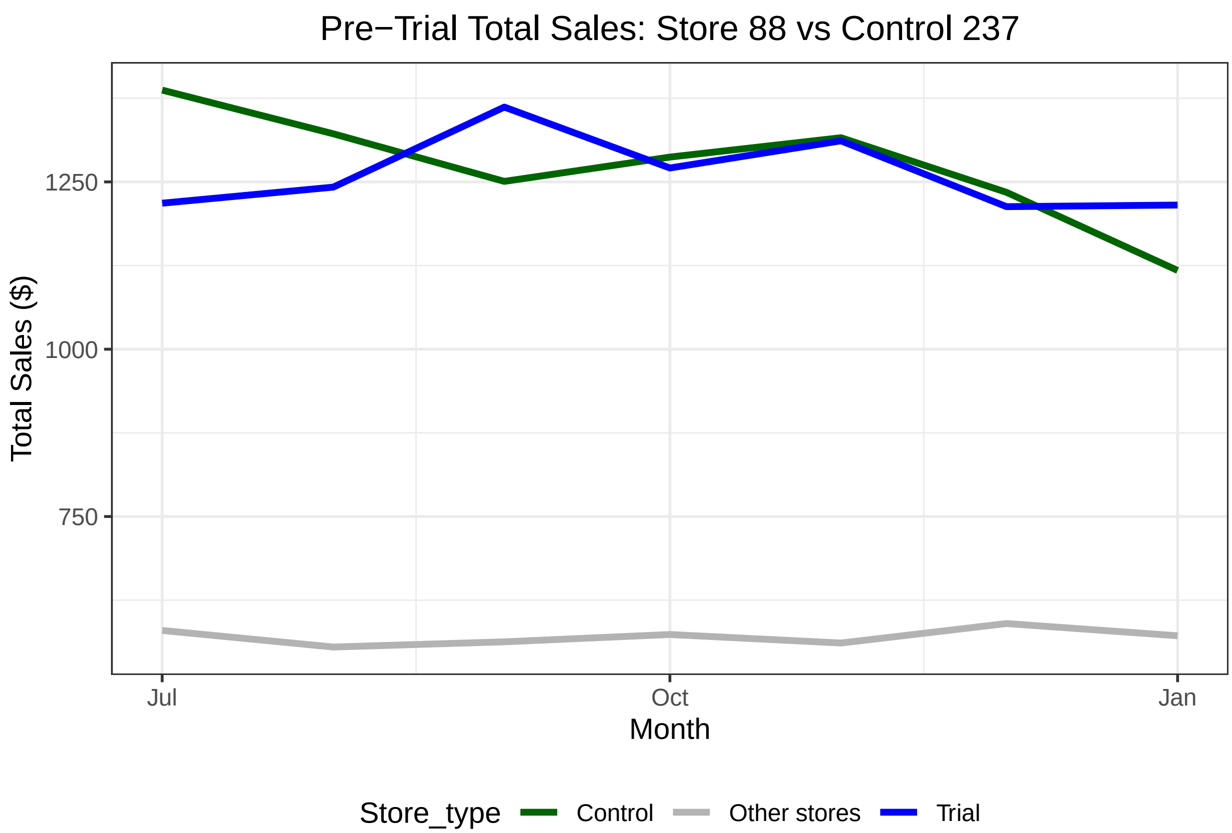

Pre−Trial Number of Customers: Store 86 vs Control 155
Number of Customers
100
90
80
70
Jul
Oct
Jan
Month
Store_type
Control
Other stores
Trial


Store 86 Total Sales vs Control 155
Shaded area: trial period (Feb–Apr 2019)
. Dashed lines: ±2 SD confidence interval
1000
750
500
250
0
Total Sales ($)
Jul 2018
Oct 2018
Jan 2019
Apr 2019
Month
Store_type
Control
Lower CI
Trial
Upper CI

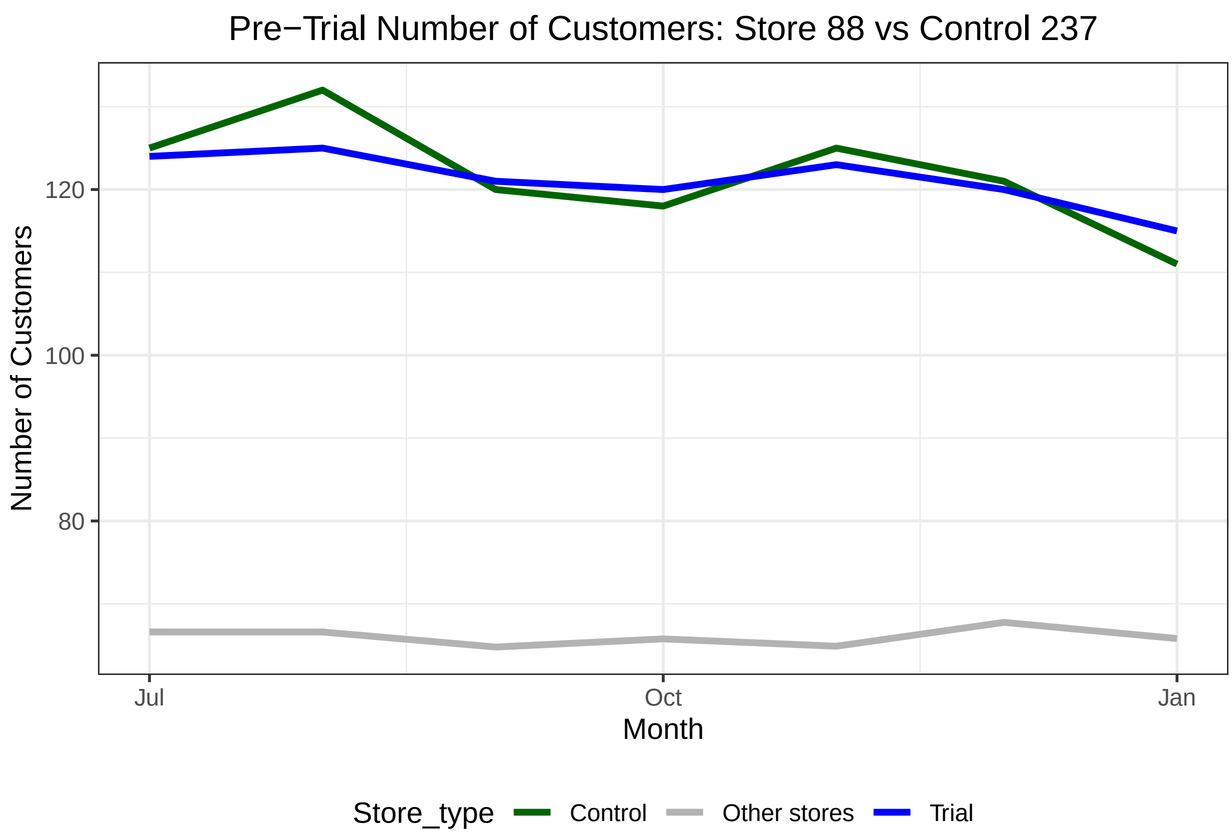


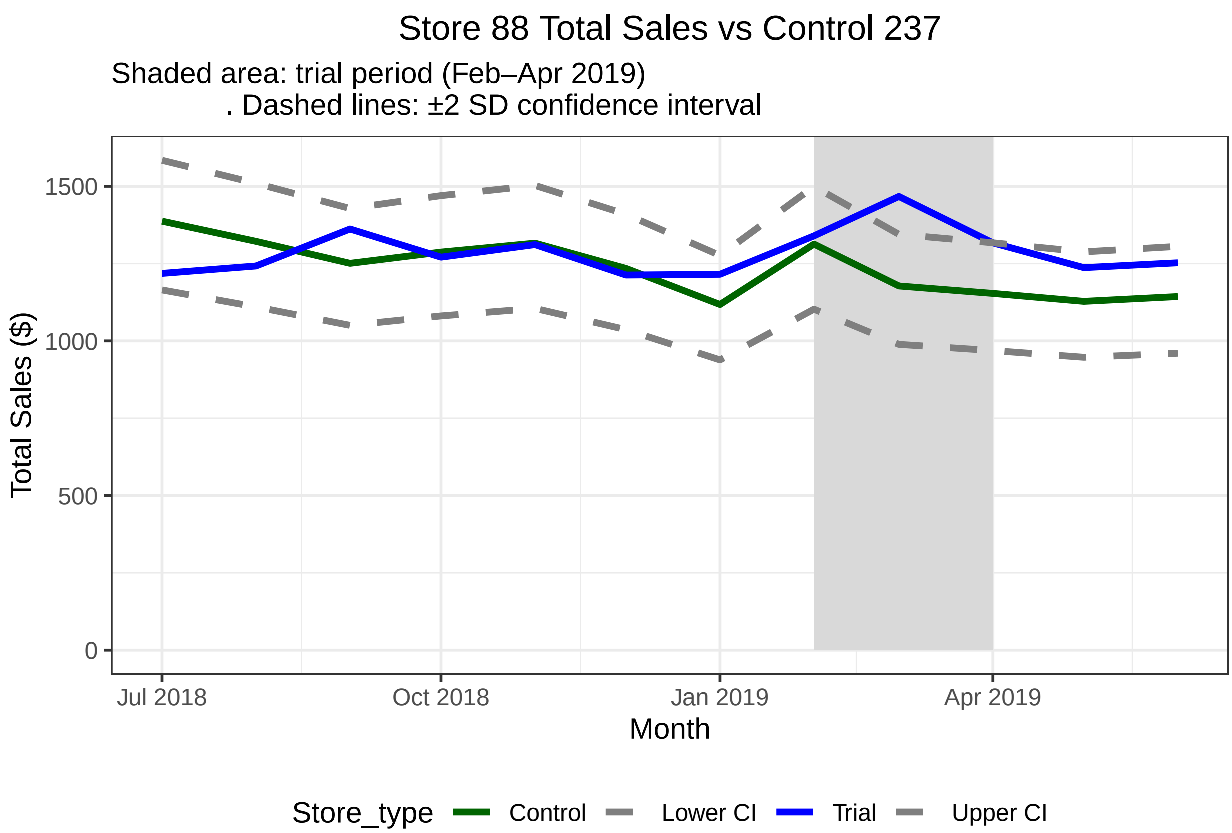

```
# ============================================================
# ANALYSIS FOR TRIAL STORE 86 vs CONTROL STORE 155
# ============================================================

# ----- TOTAL SALES UPLIFT -----
# Key: <YEARMONTH>
#   YEARMONTH    tValue Significant
 #      <num>     <num>      <lgcl>
# 1:     201902 1.3791146       FALSE
# 2:     201903 6.6777680        TRUE
# 3:     201904 0.8316513       FALSE

# ----- NUMBER OF CUSTOMERS UPLIFT -----
#Key: <YEARMONTH>
#   YEARMONTH   tValue Significant
  #      <num>    <num>      <lgcl>
# 1:     201902 4.039845        TRUE
# 2:     201903 5.369179        TRUE
# 3:     201904 1.796826       FALSE
res88 <- analyze_trial_store(88, 237)
```

```
##
## ============================================================
## ANALYSIS FOR TRIAL STORE 88 vs CONTROL STORE 237
## ============================================================
```

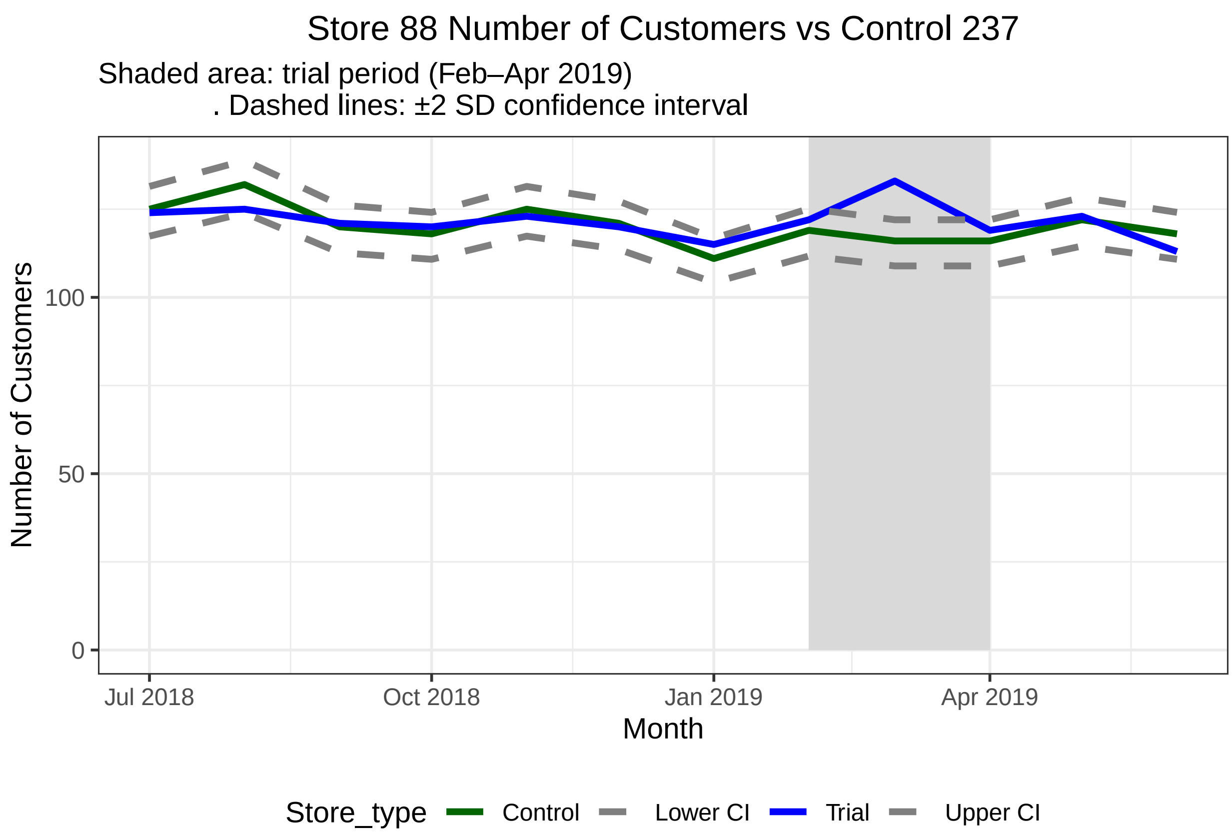



```
##
## ----- TOTAL SALES UPLIFT -----
## Key: <YEARMONTH>
##    YEARMONTH    tValue Significant
##        <num>     <num>      <lgcl>
## 1:    201902 0.3903595       FALSE
## 2:    201903 3.3751931        TRUE
## 3:    201904 1.9971030        TRUE
##
## ----- NUMBER OF CUSTOMERS UPLIFT -----
## Key: <YEARMONTH>
##    YEARMONTH   tValue Significant
##        <num>    <num>      <lgcl>
## 1:    201902 1.060206       FALSE
## 2:    201903 5.362080        TRUE
## 3:    201904 1.083321       FALSE
```


Pre−Trial Number of Customers: Store 88 vs Control 237
Number of Customers
120
100
80
Jul
Oct
Jan
Month
Store_type
Control
Other stores
Trial



Store 88 Total Sales vs Control 237
Shaded area: trial period (Feb–Apr 2019)
. Dashed lines: ±2 SD confidence interval
Total Sales ($)
1500
1000
500
0
Jul 2018
Oct 2018
Jan 2019
Apr 2019
Month
Store_type
Control
Lower CI
Trial
Upper CI




```
# ============================================================
# ANALYSIS FOR TRIAL STORE 88 vs CONTROL STORE 237
# ============================================================

# ----- TOTAL SALES UPLIFT -----
# Key: <YEARMONTH>
 #  YEARMONTH    tValue Significant
  #      <num>     <num>      <lgcl>
# 1:     201902 0.3903595       FALSE
# 2:     201903 3.3751931        TRUE
# 3:     201904 1.9971030        TRUE

# ----- NUMBER OF CUSTOMERS UPLIFT -----
# Key: <YEARMONTH>
 #  YEARMONTH   tValue Significant
  #      <num>    <num>      <lgcl>
# 1:     201902 1.060206       FALSE
# 2:     201903 5.362080        TRUE
# 3:     201904 1.083321       FALSE
```

7. SUMMARY TABLE

```r
uplift_summary <- data.table(
     Trial_Store = c(77, 86, 88),
     Control_Store = c(233, 155, 237),
     Sales_Significant = c(res77$sales_sig, res86$sales_sig, res88$sales_sig),
     Customers_Significant = c(res77$cust_sig, res86$cust_sig, res88$cust_sig))
 uplift_summary[, Recommendation := ifelse(Sales_Significant, "Rollout", "Investigate")]

cat("\n===== FINAL SUMMARY =====\n")
```

```
##
## ===== FINAL SUMMARY =====
```

```r
# ===== FINAL SUMMARY =====
print(uplift_summary)
```

```
##    Trial_Store Control_Store Sales_Significant Customers_Significant
##          <num>         <num>            <lgcl>                <lgcl>
## 1:          77           233              TRUE                  TRUE
## 2:          86           155             FALSE                  TRUE
## 3:          88           237              TRUE                 FALSE
##    Recommendation
##            <char>
## 1:        Rollout
## 2:    Investigate
## 3:        Rollout
```

```r
 #  Trial_Store Control_Store Sales_Significant Customers_Significant Recommendation
  #       <num>         <num>            <lgcl>                <lgcl>         <char>
# 1:          77           233              TRUE                  TRUE        Rollout
# 2:          86           155             FALSE                  TRUE    Investigate
# 3:          88           237              TRUE                 FALSE        Rollout

cat("\n===== RECOMMENDATION =====\n")
```

```
##
## ===== RECOMMENDATION =====
```

===== RECOMMENDATION =====

```r
cat("Store 77: Sales significant in Mar & Apr, Customers significant → ROLLOUT\n")
```

```
## Store 77: Sales significant in Mar & Apr, Customers significant → ROLLOUT
```

```r
# Store 77: Sales significant in Mar & Apr, Customers significant → ROLLOUT
cat("Store 86: Sales NOT significant (only Mar),
    but Customers significant all 3 months → INVESTIGATE (possible pricing/promotion issue)\n")
```

```
## Store 86: Sales NOT significant (only Mar),
##     but Customers significant all 3 months → INVESTIGATE (possible pricing/promotion issue)
```

```r
# Store 86: Sales NOT significant (only Mar),
# but Customers significant all 3 months → INVESTIGATE (possible pricing/promotion issue)
cat("Store 88: Sales significant in Mar & Apr, Customers significant → ROLLOUT\n")
```

```
## Store 88: Sales significant in Mar & Apr, Customers significant → ROLLOUT
```

```r
# Store 88: Sales significant in Mar & Apr, Customers significant → ROLLOUT
cat("\nRecommend rolling out the new layout to stores similar to 77 and 88.\n")
```

```
##
## Recommend rolling out the new layout to stores similar to 77 and 88.
```

```r
# Recommend rolling out the new layout to stores similar to 77 and 88.
```